

Debarghya Roy

(847) 951-0902 | debroy2021@gmail.com | U.S. Citizen | Buffalo Grove, IL | linkedin.com/in/debroy2024

EDUCATION

University of Illinois at Urbana Champaign (UIUC)

Aug. 2025 – Expected May 2029

Bachelor of Science in Electrical Engineering

Champaign, IL

- Cumulative GPA: 3.88 / 4.00, Honors: James Scholar, Dean's List
- Relevant Coursework: ECE 310 – Digital Signal Processing, ECE 210 – Analog Signal Processing, PHYS 214 – Quantum Physics, PHYS 213 – Thermal Physics, MATH 257 – Linear Algebra, MATH 285 – Intro to Differential Equations

EXPERIENCE

Eco Illini Supermileage

Aug. 2025 – Present

Driver Display, Telemetry & Testing/Characterization Lead

Champaign, IL

- Developed a custom ESP32-based telemetry and driver display system with CAN, Wi-Fi, data logging, and sensors.
- Using MATLAB Simulink to model vehicle performance, supporting efficiency optimization and race strategy development.
- Participated in the 2026 Shell Eco Marathon Competition and achieved an efficiency of 186.5 km/kWh.
- Engineered a 97%-efficient buck converter (32–42V to 12V @ 5A) as part of an electric vehicle's Power Distribution Unit.

ECE 145: First Year Design Laboratory

Aug. 2026 – Present

Classroom Assistant

- Mentoring freshmen through semester-long honors projects, providing guidance on design, prototyping, and debugging.
- Supporting students with hands-on troubleshooting using oscilloscopes, multimeters, and other electronics lab equipment.

Illinois MicroTech

Oct. 2025 – May 2026

Optics and Electronics Team Member

Champaign, IL

- Member of the CMU based Hackerfab project, a student led initiative created to make nanofabrication more accessible.
- Strategized and implemented solutions to resolve key operational challenges with the Lithography Stepper.
- Investigated and mitigated UV lithography imaging artifacts and experimentally determined optical resolution and feature size accuracy down to 0.2um/pixel.

Science Reach, Adlai E. Stevenson High School

Aug. 2023 – May 2025

Researcher

Lincolnshire, IL

- Authored “Comparative Analysis of CNNs for Classifying Alzheimer’s Disease.”
- Achieved 95% accuracy using VGG16 model with the Convolutional Block Attention Mechanism.
- Improved early detection of Alzheimer’s for more than 55 million patients with dementia.
- Earned the Jay and Jeremy Hoffman Memorial Award and Gold Honors at Illinois Junior Academy of Sciences State Science Fair.

Parabole.ai

May 2024 – Aug. 2024

Intern

Remote

- Led a team of four interns to migrate the company’s website from Flask to React, improving site responsiveness.
- Integrated the company’s TRAIN platform API to generate causal inference graphs, enhancing usability for 100+ users.
- Collaborated with project leaders to deliver production-ready features in 2 months.

SERVICE & PROJECTS

ECE 145: First Year Design Laboratory

Aug. 2025 – Dec. 2025

Honors Project

Champaign, IL

- Created a remote-controlled, multi-colored, automatic lighting system that adjusts to ambient light using analog circuitry.
- Maintained a detailed journal and project proposal to document the whole design and development process.

The Parris Foundation: S.T.E.M.ulating Minds

Aug. 2022 – Jul. 2026

Volunteer

Remote

- Tutored underprivileged 6–12th grade students in Computer Science, Python, and Mathematics on a weekly basis.
- Fostered academic growth by providing personalized guidance and support to students.
- Recognized with 4 awards for Volunteer and Tutor dedication.

TECHNICAL SKILLS

- **Languages & Libraries:** C, ReactJS, Java, Python, TensorFlow, Scikit Learn, Pandas, NumPy, Unix/Linux Shell
- **Tools:** KiCad, Raspberry Pi, STM32CubeIDE, Oscilloscopes, Function Generators, Git